Ambient WiFi as a Cheap Radar for Automotive SLAM: Centimetre Localization, a Phantom Ceiling on Mapping, and What That Ceiling Is — in Simulation and on $30 of Silicon

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***Abstract*—We ask whether ambient sub-7 GHz WiFi, received on a moving vehicle and processed with commodity channel state information (CSI), can serve as a low-cost radar/LiDAR replacement for simultaneous localization and mapping (SLAM). Using a physics-based, ray-traced (Sionna RT) substrate and, for the first time in this line of work, a $30 ESP32 hardware testbed, we find a consistent split. *Localization* is centimetre-level and, we show, structurally robust: because the estimator is odometry-anchored and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point–position error up to tens of metres and to sudden, total measurement blackouts. *Mapping*, by contrast, hits a hard ceiling: with realistic commodity CSI, most detections are phantoms (≈89%) accompanied by a several-metre range bias — a front-end and geometry limit, not a path-discrimination or resolution limit, and one a learned filter cannot repair. Comparing the same substrate against real LiDAR (KITTI) and a simulated 77 GHz FMCW radar shows the ceiling is neither carrier-specific nor WiFi-specific: the radio carrier does not matter, geometry does (a monostatic on-vehicle transmitter nearly eliminates the phantoms), and wider bandwidth makes them worse. Finally, we reproduce the delay-domain reading on real ESP32 silicon. The practical message: WiFi localization rivals far costlier sensors, while WiFi mapping is capped by a front-end/geometry phantom ceiling that we characterize and, in part, show how to escape.**

***Index Terms*—WiFi sensing, channel state information, SLAM, passive radar, integrated sensing and communication, automotive, ESP32, low-cost hardware, multipath.**

I. Introduction

THE exteroceptive sensors that anchor vehicle SLAM — LiDAR and radar — are also the costly part of the perception stack. A research-grade spinning LiDAR runs into the thousands of dollars and an automotive radar into the tens to hundreds, while the WiFi silicon a vehicle already carries costs tens. If ambient WiFi — signals radiated anyway, by infrastructure that already exists, received on an antenna already on the vehicle — could stand in for those sensors, the sensing bill for autonomy would change by orders of magnitude. This paper asks the question without hedging: *can ambient sub-7 GHz WiFi replace radar or LiDAR as a SLAM front-end?*

The honest answer is neither "yes" nor "no" but a boundary, and the boundary is the contribution. WiFi replaces the costly sensors for one half of SLAM — localization — and fails at the other — mapping — and the failure is not where the literature, including our own earlier feasibility study [1], assumed it was. We locate that failure precisely, show it is *not specific to WiFi*, and then reproduce the underlying reading on real hardware.

Localization is centimetre-level and, we show, *structurally* robust: because the estimator is anchored by vehicle odometry and the bistatic CSI measurements are refinement-only, absolute trajectory error is insensitive to access-point (AP) position error up to tens of metres and to sudden, total measurement blackouts. *Mapping* hits a hard ceiling: with realistic commodity CSI, the overwhelming majority of detections (≈89%) are *phantoms* — they match no real propagation path — accompanied by a several-metre range bias. This is a front-end and geometry limit, not a path-discrimination or resolution limit, and a learned filter cannot repair it, because one cannot select real paths from a set in which most detections are not real paths.

To test whether this ceiling is a property of WiFi or of RF sensing more broadly, we run the *same* substrate against real LiDAR (KITTI [2]) and a simulated 77 GHz FMCW radar. The result is decisive: the radio carrier does not matter, geometry does — a *monostatic*, on-vehicle transmitter nearly eliminates the phantoms — and wider bandwidth makes them *worse*. Finally, licensed by that carrier-independence, we move the claim off simulation entirely and capture, parse, and delay-profile real HT40 CSI on two $30 ESP32-S3 boards.

Contributions

- A physics-based feasibility study of on-vehicle, mobile, outdoor WiFi as a SLAM front-end — an unoccupied point in the design space (prior passive-WiFi-

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radar is indoor/static/infrastructure-fixed; the one moving-platform passive radar uses 5G).

- Identification and mechanism of the *mapping phantom ceiling* ($\approx$89% phantom detections + range bias), and proof it is not a discrimination or resolution limit and not repairable by a learned filter.
- Evidence the ceiling is *universal*: carrier-independent and not WiFi-specific, via a controlled comparison against LiDAR and a 77 GHz radar on one substrate; geometry (monostatic vs bistatic), not carrier or bandwidth, is decisive.
- A head-to-head WiFi-versus-LiDAR comparison on identical scenes with a KITTI-validated LiDAR back-end, plus a *sensor-cost* analysis (the WiFi package is 84–600$\times$ cheaper) and cost-normalized accuracy (Secs. VI–VII).
- A symmetric fusion study with a design rule: fusion reinforces at sensor parity and *degrades* the stronger sensor by $8\times$ when badly mismatched (Sec. VIII).
- A robustness characterization: localization is structurally tolerant of access-point–position error and measurement blackouts (Sec. IX-E).
- The first delay-resolved reflector reading from an ESP32 on \$30 of hardware, moving the claim off simulation (Sec. X).

II. Related Work

WiFi/CSI sensing is mature but overwhelmingly *indoor, static, and infrastructure-fixed*. Through-the-wall and device-free sensing [3], [4], [5], CSI-based ranging and imaging [6], [7], [8], and pose or point-cloud recovery [9], [10] all assume fixed transmitters and a static or slowly-varying scene. None places the sensing platform on a moving vehicle outdoors.

WiFi-based SLAM takes two forms. Channel- and multipath-based SLAM treats reflectors as virtual anchors [11], [12], [13], [14], and radio-fingerprint or P2SLAM approaches localize against known access points [15], [16] — both *infrastructure-bound*: they need the transmitters' positions and are demonstrated indoors. The alternative uses WiFi only to *assist* a primary ranging sensor (LiDAR or IMU) [17], [18], [19], not to replace it. No prior work evaluates WiFi as a stand-alone, on-vehicle, outdoor *replacement* front-end.

Integrated sensing and communication (ISAC/JCAS) envisions the same waveform serving communication and sensing [20], [21], [22], [23], but its SLAM instantiations are early [24]. At mmWave, 60 GHz 802.11ad has been repurposed as a radar [25], [26]; the only *moving-platform* passive radar we are aware of uses 5G, not WiFi [27].

The gap. On-vehicle, mobile, outdoor WiFi (802.11) as a radar/LiDAR replacement for SLAM is an unoccupied point in this design space. This paper occupies it, and — unlike the infrastructure-bound line above — shows that the strongest configuration (monostatic, on-vehicle transmitter) needs *no* known AP positions at all.

III. Method: One Substrate, Three Sensors, One Testbed

A single substrate — identical scenes, trajectories, footprint ground truth, and metric suite — carries WiFi, LiDAR, and 77 GHz radar, so differences are attributable to the sensor, not the harness. The same detection chain then runs on real ESP32 hardware.

A. Bistatic geometry

A vehicle carrying a receive antenna array drives through an outdoor scene illuminated by N_{AP} ambient WiFi access points at known positions. Each specular component follows an access point $\rightarrow$ reflector $\rightarrow$ vehicle path, so the measured path length $\rho = \|\mathbf{p}_{\mathrm{AP}} - \mathbf{r}\| + \|\mathbf{r} - \mathbf{p}_{\mathrm{veh}}\|$ places the reflector $\mathbf{r}$ on an ellipse with foci at the AP and the vehicle. Given the arrival azimuth ϕ, the reflector range s along $\hat{\mathbf{u}}(\phi)$ solves

$$s = \frac{\rho^2 - d_{\mathrm{AP}}^2}{2\big(\rho - \mathbf{d}_{\mathrm{AP}}\cdot\hat{\mathbf{u}}(\phi)\big)}, \qquad \mathbf{d}_{\mathrm{AP}} = \mathbf{p}_{\mathrm{AP}} - \mathbf{p}_{\mathrm{veh}}, \qquad (1)$$

with $d_{\mathrm{AP}} = \|\mathbf{d}_{\mathrm{AP}}\|$; direct-path ($s \leq 0$) and implausible solves are rejected. A *monostatic* variant places the transmitter on the vehicle itself ($\mathbf{p}_{\mathrm{AP}} \rightarrow \mathbf{p}_{\mathrm{veh}}$): the measurement becomes a round-trip range, and (1) needs no external AP position at all — a fact that turns out to decide both the phantom rate (Sec. IX) and the AP-error robustness (Sec. IX-E).

B. Ray-traced channel and scenes

Channels are generated with Sionna RT 2.0 (Mitsuba/Dr.Jit), which ray-traces the scene to produce per-path delays, angles, and interaction types and the frequency-domain channel across N_{sc} subcarriers and N_{rx} antennas per vehicle pose; additive white Gaussian noise is applied at a configured SNR. Radar is electromagnetic, so the same tracer models it natively; LiDAR requires an optical (diffuse-scattering) treatment (Sec. III-E).

C. CSI front-end: joint 2-D (delay–angle) MUSIC

We compare two sensing tiers. The *oracle* front-end reads Sionna's true single-scatter path parameters, isolating the geometry from estimation error. The *realistic* front-end estimates delay and azimuth from noisy CSI with MUSIC. A key refinement is *joint 2-D (delay–angle) MUSIC* with 2-D spatial smoothing (SVD subspace; delay grid bounded to the unambiguous $1/\Delta f$ range), which recovers each path's delay and angle *together*, so association is intrinsic rather than paired by sorted index. Array-relative electrical angles map to world azimuth by $\sin\beta = -\sin\theta$ for the lateral ULA.

D. Bistatic-ellipse mapping and particle-filter SLAM

A Rao–Blackwellized particle filter propagates the vehicle pose from odometry and weights particles by the bistatic consistency (1) of each detection; triangulated reflectors are accumulated and consensus-filtered into a landmark map. We report trajectory error (ATE, RPE) and, for the map, symmetric Chamfer distance, occupancy IoU, and *directional* accuracy and completeness (a passive-WiFi map illuminates only a subset of surfaces). Map ground truth is each scene object's facade footprint.

E. Complexity and real-time feasibility

The 2-D MUSIC front-end is the cost driver: per AP per frame it forms a spatially-smoothed covariance over the delay $\times$ angle grid and takes its eigendecomposition, an $\mathcal{O}(N^3)$ operation in the smoothed dimension N. Two facts keep this practical. First, N is small (a bounded delay grid and a single ULA), and the work is per-AP-per-frame and embarrassingly parallel. Second — and more consequentially — Sec. IX shows that MUSIC is precisely what *creates* the mapping phantoms, and that a calibrated constant-false-alarm-rate (CFAR) detector in the monostatic geometry both removes most phantoms *and* is far cheaper: the monostatic delay-domain chain we port to hardware costs $\approx$194 kflop and $\approx$26 KB per sweep, which an ESP32-S3 executes with $\approx$250$\times$ headroom. Real-time on-vehicle execution and the mapping fix therefore point the same way.

F. LiDAR and 77 GHz radar baselines

The LiDAR baseline is an Ouster-class spinning sensor, ray-traced with a diffuse-scattering model and a scan-to-map ICP back-end that is *externally validated* on real KITTI sequences ($\approx$0.3% drift), so its simulated numbers are anchored to reality. The radar baseline is a 77 GHz FMCW sensor with a range–azimuth CFAR chain. One honest caveat propagates from the radar study: our point-to-point ICP back-end cannot recover *rotation* from spinning-radar point clouds (the yaw cost is flat), so the radar/WiFi mapping ablation (Table IV) is scored under *ground-truth poses* — it isolates the front-end and geometry, which is exactly the quantity of interest.

G. ESP32-S3 hardware testbed

Two $30 ESP32-S3 boards realise the monostatic geometry directly: one illuminates (HT40, 40 MHz) and one logs CSI $\approx$0.5 m away, so the round-trip delay taps *are* echo ranges. Only the *excess* delay (relative to the line-of-sight arrival) is needed, and that is preserved on commodity CSI even though absolute time-of-flight is not; the 0.5 m baseline is under 7% of one 40 MHz range cell, so the rig is monostatic to within measurement. Using 2.4 GHz hardware is licensed by the carrier-independence result (Table IV, B$\rightarrow$C). Details in Sec. X.

IV. Experimental Setup

Two outdoor scenes are used throughout: a *controlled* scene (an isolated reflecting wall, which isolates the geometry) and a *street canyon* (dense, multi-surface, adversarial for multipath). A vehicle follows a 240-frame trajectory through each, illuminated by three ambient access points. Unless stated otherwise the WiFi link is 5.2 GHz with 40 MHz of bandwidth, 64 subcarriers, a four-element half-wavelength ULA, and 20 dB SNR; the bandwidth sweep in Sec. V varies this from 20 to 160 MHz. Channels come from Sionna RT 2.0. Stochastic results are reported as mean $\pm$ standard deviation over five seeds. Ground truth for the map is each scene object's facade footprint; trajectory ground truth is the simulated pose. All configurations, seeds, and the released dataset are in the artifact accompanying this paper.

V. Feasibility: Localization Works, Mapping Is Two-Tier

A. Localization operating envelope

Ambient WiFi localizes the vehicle to centimetre level. Sweeping bandwidth shows the expected monotonic $\sim$20$\times$ ATE improvement, from 0.78 m at 20 MHz to 0.038 m at 160 MHz (0.093 m at the nominal 40 MHz), tracking the range resolution $\Delta R = c/2B$; SNR gives the same clean trend (2.02 m at 0 dB to 0.076 m at 30 dB), and more access points and higher speeds help. Localization has a clean operating envelope — it works wherever the link budget does.

B. Realistic sensing: joint delay–angle MUSIC

Separate 1-D delay and 1-D angle estimates paired by sorted index mis-associate in multipath and scatter phantom reflectors: this naive front-end reaches only 0.73 m ATE. Joint 2-D MUSIC, which recovers each path's delay and angle together, lifts realistic-CSI localization $\sim$7$\times$ to $\mathbf{0.098 \pm 0.028}$ m (five seeds, controlled scene, 160 MHz) — *approaching* but not matching the oracle-sensing 0.049 ± 0.027 m. So realistic commodity CSI localizes at a few centimetres, close to the information limit of the geometry.

C. Mapping is two-tier

Mapping splits sharply by front-end. With *oracle* single-bounce sensing the bistatic back-end reconstructs reflecting surfaces to $\approx$25–30 cm (controlled: 0.25 m accuracy, 0.79 occupancy IoU; street: 0.30 m) — the geometry is sound. With *realistic* commodity CSI, map accuracy collapses to $\approx$4.8 m. The same estimator that localizes to centimetres maps to metres.

D. It is not a bandwidth or aperture limit

The obvious hypothesis — insufficient resolution — is wrong. Repeating the realistic-CSI map at 60 GHz with 1.76 GHz of bandwidth, and again with a 16-antenna array, leaves the $\approx$4.8 m floor essentially unchanged: neither an order of magnitude more bandwidth nor a larger aperture moves it. The mapping ceiling is therefore a property of the *front-end*, not of resolution — which motivates asking exactly *what* the front-end does wrong (Sec. IX).

VI. Accuracy: WiFi versus LiDAR

Feasibility in isolation is not a replacement claim; the question is how WiFi compares with the sensor it would replace, on the same scenes and metrics. We simulate two LiDAR configurations — LiDAR-A (a mid-range spinning unit) and LiDAR-B (a lower-grade variant) — with the back-end externally validated on KITTI (Sec. IX-C). Table I is the head-to-head.

On localization, WiFi reaches parity with LiDAR. In the controlled scene realistic WiFi achieves 0.098 m ATE against LiDAR-A's 0.102 m — a statistical tie — and is comfortably better than LiDAR-B (0.483 m). This is the central positive result. In the street canyon the denser multipath costs WiFi

TABLE I
WIFI VERSUS LIDAR ON IDENTICAL SCENES AND METRICS (M; LOWER IS BETTER EXCEPT IOU). WIFI TIES LIDAR-A ON LOCALIZATION AND SCORES ZERO ON MAPPING.

Scene	Sensor	ATE	Chamfer	m-acc	IoU
Controlled	WiFi oracle	0.049 ± 0.027	0.505	0.248	0.791
	WiFi realistic	0.098 ± 0.028	5.974	2.490	0.000
	LiDAR-A	0.102	0.209	0.250	0.977
	LiDAR-B	0.483	0.187	0.251	**1.000**
Street	WiFi oracle	0.104 ± 0.041	12.344	0.309	0.077
	WiFi realistic	0.314 ± 0.051	18.086	3.525	0.001
	LiDAR-A	**0.026**	8.674	0.251	0.163
	LiDAR-B	0.857	3.734	2.125	**0.261**

an order of magnitude (0.314 m versus LiDAR-A's 0.026 m), so the parity is scene-dependent: WiFi localization degrades gracefully where LiDAR does not.

On mapping, WiFi does not merely lose — it scores zero. LiDAR reaches occupancy IoU up to 1.00; realistic WiFi reaches ≈ 0 on both scenes. That oracle-sensing WiFi *does* map (IoU 0.791 on the controlled wall) localizes the failure squarely in the front-end, which Sec. IX then dissects.

VII. COST

The replacement question is ultimately economic, so we price it, reporting dated ranges rather than point estimates. The vehicle-side WiFi sensing package — a commodity CSI receiver [28] plus antennas — is \$40–95; an ESP32-class node [29] is \$10–35. LiDAR spans four orders of magnitude: legacy spinning (\$75,000–80,000), the mid-range spinning class we simulate (\$8,000–24,000), emerging automotive solid-state (\$100–600), and a budget 2-D scanner [30] (\$99) — the last a *price floor*, not an automotive peer, and never compared on mapping quality.

The gap. Against the LiDAR class we actually simulate, the WiFi package is **84–600**$\times$ cheaper; against legacy spinning, 789–2,000$\times$.

Cost-normalized performance sharpens it. On localization value (price $\times$ ATE, lower better) WiFi delivers 4–30 \$·m against LiDAR's 208–2,448 \$·m — one to two orders of magnitude more accuracy per dollar. The gain comes from price, not precision: WiFi *matches* LiDAR-A's accuracy (Sec. VI) at two orders of magnitude less cost. On mapping value (price / IoU) WiFi's cost is *infinite* — it buys no map coverage at any price. LiDAR is the only modality here that converts money into map.

The claim is conditional, and we state the condition. It rests on the ambient-AP premise. If a deployment must *install* its own access points, the WiFi package rises to \$130–335 for three APs and can become *more expensive* (0.3–1.5$\times$) than the cheapest solid-state LiDAR. The dramatic cost story is a story about infrastructure that already exists, and should be quoted that way — another reason the monostatic, on-vehicle geometry (Sec. IX-D) is attractive: it needs no external infrastructure at all.

TABLE II
FUSION ATE (M). FUSION REINFORCES WHEN THE SENSORS ARE OF COMPARABLE QUALITY AND *degrades* THE STRONGER ONE WHEN THEY ARE BADLY MISMATCHED.

Scene	LiDAR cfg	WiFi	LiDAR	Fusion
Controlled	A	0.081	0.102	**0.065**
Controlled	B	0.081	0.212	**0.044**
Street	A	0.281	**0.027**	0.218
Street	B	0.281	0.844	**0.175**

VIII. FUSION: DOES KEEPING BOTH PAY?

If WiFi is nearly free, the natural question is whether to fuse rather than replace. We fuse the two measurement streams symmetrically in the same back-end (Table II).

Fusion helps, conditionally. Against LiDAR-B it improves ATE by 79% on both scenes — reinforcement, not mere addition. **And it can hurt:** in the street/LiDAR-A configuration the LiDAR alone achieves 0.027 m and fusion degrades it to 0.218 m, an $8\times$ regression. **The condition is sensor parity** — fusion reinforces when the two sensors are of comparable quality and corrupts the stronger when they are not, which is a design rule, not a tuning artefact. On the map, the union adds coverage exactly where LiDAR is incomplete (street IoU $0.149 \rightarrow 0.177$ for A; $0.262 \rightarrow 0.309$ for B) while mildly polluting an already-good map (controlled $0.977 \rightarrow 0.913$).

Is it worth paying for both? The WiFi package is a 0.2–1.2% addition to a LiDAR budget — roughly half a percent more money for a 36–79% better trajectory when the sensors are matched. That is the strongest practical case for WiFi in this paper: not as a replacement for a good LiDAR, but as an almost-free reinforcement of a modest one.

IX. THE MAPPING CEILING AND ITS MECHANISM

The mapping failure is the central puzzle: under oracle sensing the same geometry maps to 0.25 m, yet under realistic CSI it collapses to occupancy IoU ≈ 0. Something in the front-end destroys the map. Locating it precisely — and identifying *which* of the plausible culprits actually dominates — is this section's contribution, and it also answers directly when a reflector detection can and cannot be promoted to a landmark.

A. A learned discriminator cannot fix it

The feasibility study attributed the floor to *path discrimination* (telling a genuine facade reflection from LOS, floor, and multi-bounce returns) and argued that this label is learnable. The natural next step is to close the loop: put the discriminator *inside* the mapping pipeline. It does not help. A ladder of filters over the MUSIC detections — none, a physics heuristic, a random forest, and a small neural network — gating which detections enter the map while leaving the particle weights (and hence localization) untouched, *leaves the occupancy IoU at* 0.000: the learned rungs reject nearly everything and empty the map. Trained only on features a single-ULA delay–azimuth front-end can actually measure, the discriminator's held-out F1 is 0.00–0.45. The earlier optimism came from a feature (arrival *elevation*) that such a front-end never estimates, and

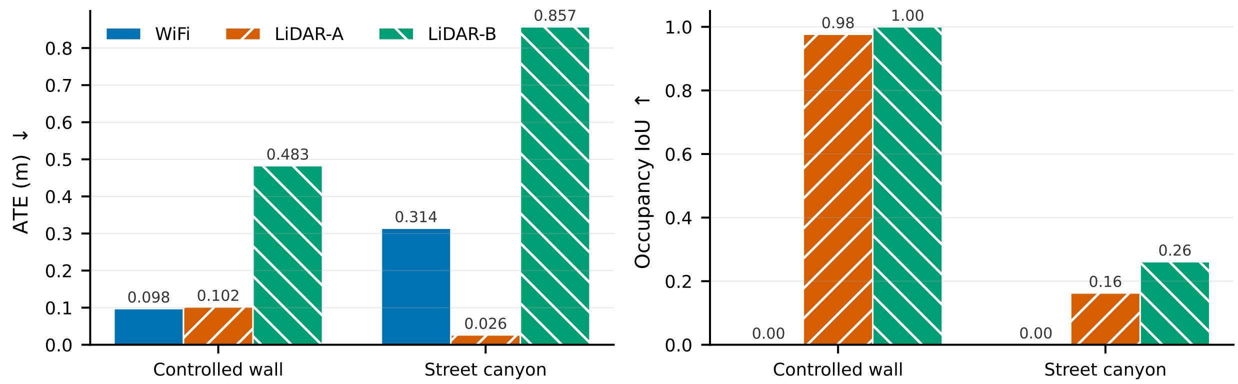

Fig. 1. WiFi wins localization and loses mapping outright. Realistic commodity-CSI WiFi ties the mid-range LiDAR on trajectory error in the controlled scene, while its occupancy IoU is essentially zero on both scenes.

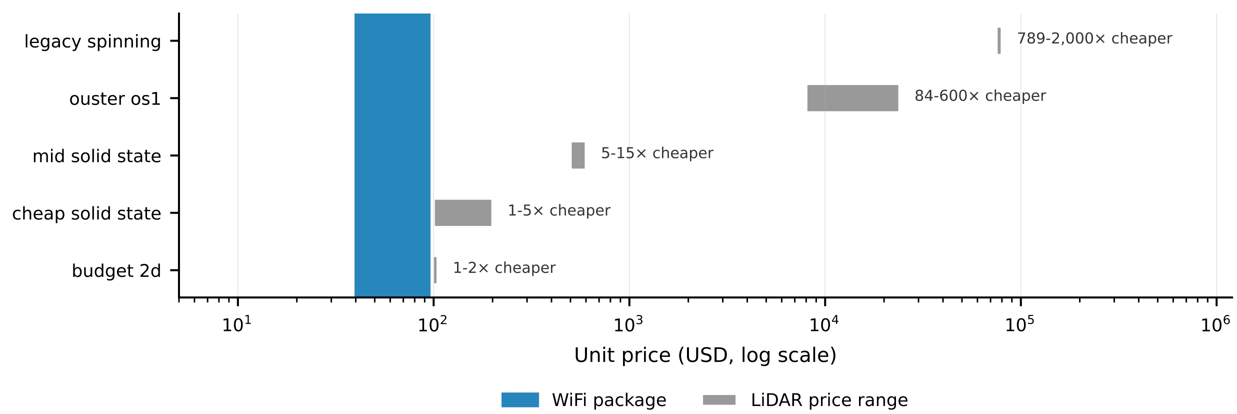

Fig. 2. The sensor-cost envelope (log scale). The WiFi package is 84–600× cheaper than the mid-range spinning LiDAR whose parameters our models simulate. The budget 2-D scanner is shown as a price floor, not as an automotive peer.

from classifying *true ray-traced paths* rather than the *MUSIC detections* the deployed system must classify — which, as we now show, are not the same objects.

B. Isolating the cause

We match every MUSIC detection to its nearest true path, and for those matching a genuine single-bounce facade path we triangulate the same path twice — once with the ray tracer's true delay and azimuth, once with MUSIC's estimates — so any degradation is *pure estimation error with discrimination held perfect* (Table III). Three mechanisms emerge, in order of importance. **(1) Phantoms (≈ 89%) dominate:** nearly nine in ten MUSIC detections match *no real propagation path at all* — they are estimator artefacts. One cannot select the real paths from a set in which most detections are not real paths, which is exactly why the discriminator fails. **(2) Range bias (6.45 m median, controlled):** even correctly-identified facade paths are mislocated — true parameters place 100% of them within 1 m, MUSIC's place 2.4% — a *bias*, not a resolution bound, so bandwidth does not fix it (Sec. V-D). **(3) Discrimination (2–8%)** — the previously assumed culprit — is the *smallest* term. A filter selects; it can neither invent the real paths 89% of detections lack nor correct the bias, so a valid landmark track cannot be sustained under realistic CSI however the promotion threshold is tuned.

TABLE III
ISOLATING THE MAPPING CEILING. DISCRIMINATION — THE PREVIOUSLY ASSUMED CULPRIT — IS THE SMALLEST OF THREE MECHANISMS.

	Controlled	Street
Phantom (matches no real path)	**89.2%**	**89.5%**
Non-facade real path (*discrimination*)	2.2%	8.4%
True facade path	8.6%	2.0%
Triangulated < 1 m, *true* params	100%	100%
Triangulated < 1 m, *MUSIC* params	**2.4%**	76.7%
MUSIC range error (median)	6.45 m	2.11 m

C. An external anchor

Lest the whole comparison be a simulation artefact, the LiDAR back-end that shares this substrate is validated on real KITTI data at 1.16 m ATE over 394 m (≈0.3% drift), matching published LiDAR-odometry performance. The substrate is anchored to reality on at least one arm.

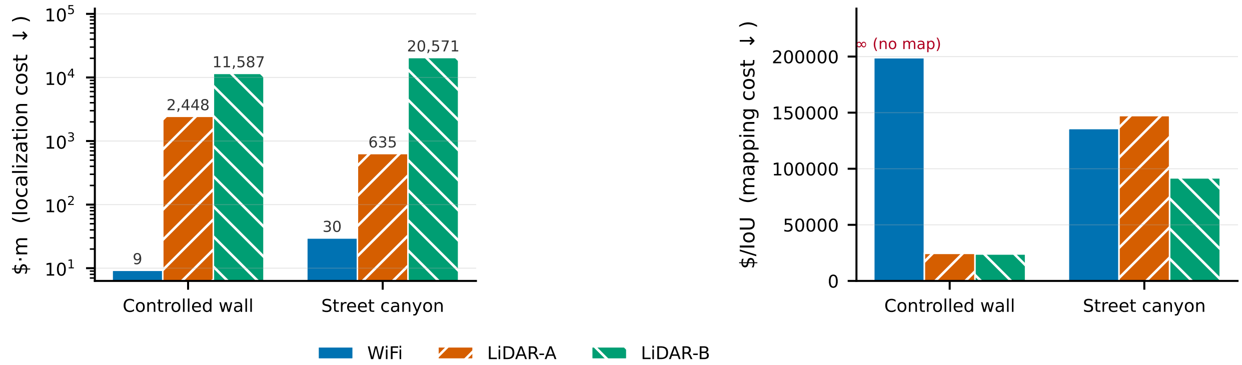

Fig. 3. Cost read together with accuracy. Left: localization cost (price × ATE) — WiFi is one to two orders of magnitude better per dollar. Right: mapping cost (price per unit IoU) — WiFi buys no map coverage at any price.

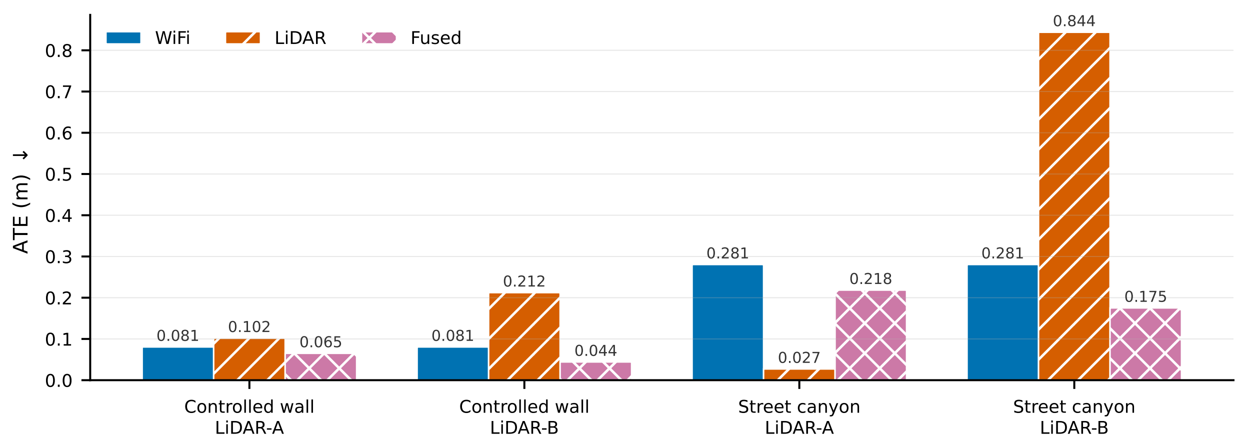

Fig. 4. Fusion is conditional. Tight fusion beats both solo sensors in three of four configurations, but in street/LiDAR-A — where the sensors are badly mismatched — it degrades the stronger sensor by $8\times$.

D. Universality: the ceiling is carrier-independent, geometry sets it

Is the ceiling a property of WiFi, or of RF sensing on limited data? We run the same substrate against a 77 GHz FMCW radar and ablate the front-end, geometry, carrier, and bandwidth independently (Table IV). Three readings stand out. **Carrier does nothing** (B→C: 5.2 GHz and 77 GHz give the same $\approx 0\%$) — which licenses building the hardware testbed at 2.4 GHz. **Geometry is everything** (A→B: moving the transmitter from off-vehicle to on-vehicle drops the phantom rate from 18.2% to 0.1%, a $180\times$ reduction). **More bandwidth is worse** (C→D: 4 GHz gives 9.0% where 160 MHz gave 0.0% — more resolvable cells, more false alarms). And MUSIC (cell M) reproduces the total mapping failure (IoU 0.003) even under ground-truth poses: the ceiling is the *front-end plus geometry*, not WiFi and not RF sensing.

A caveat we state plainly: our point-to-point ICP back-end cannot recover *rotation* from spinning-radar point clouds (the yaw cost is flat), so Table IV is scored under ground-truth poses. That is a feature here, not a bug — it isolates the front-end and geometry, which is exactly what the phantom rate measures.

TABLE IV
ABLATION ON ONE SUBSTRATE (3 SEEDS), SCORED UNDER GROUND-TRUTH POSES. GEOMETRY, NOT CARRIER OR BANDWIDTH, SETS THE PHANTOM RATE.

Cell	Front-end / geometry	Carrier / BW	Phantom rate
M	MUSIC, bistatic	5.2 GHz / 160 MHz	— (IoU 0.003)
A	CFAR, bistatic	5.2 GHz / 160 MHz	$18.2 \pm 0.6\%$
B	CFAR, monostatic	5.2 GHz / 160 MHz	$\mathbf{0.1} \pm \mathbf{0.2}\%$
C	CFAR, monostatic	77 GHz / 160 MHz	0.0%
D	CFAR, monostatic	77 GHz / **4 GHz**	$9.0 \pm 1.5\%$

E. Robustness to AP-position error and measurement loss

Two concerns follow directly from the bistatic geometry: the ellipse (1) depends on AP position, and a moving vehicle will suffer occlusions. We quantify both by perturbing the estimator on the nominal scene (eight and four seeds respectively).

AP-position error does not reach the trajectory (Fig. 6). Absolute trajectory error is flat within seed noise for AP errors up to 32 m — larger than the scene — in both the oracle and dense-MUSIC regimes. The robustness is *structural*: the particle filter requires a motion model, so it is always odometry-anchored, and the bistatic measurements are

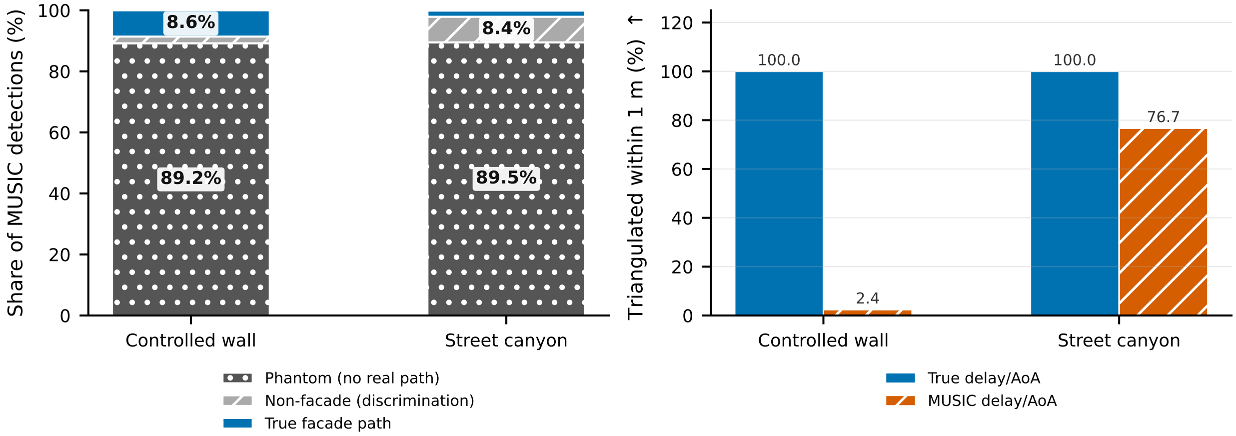

Fig. 5. The mapping ceiling. Left: where MUSIC detections come from — ≈89% correspond to no real propagation path. Right: the same genuine facade paths triangulated with true versus MUSIC parameters, isolating estimation error from discrimination.

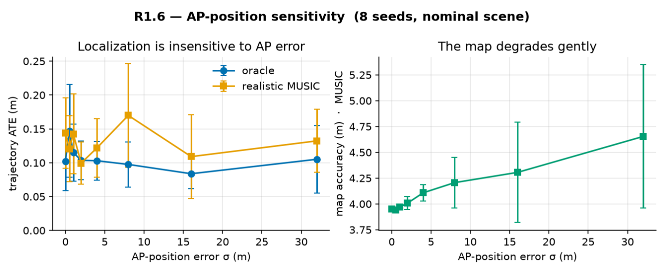

Fig. 6. AP-position sensitivity. Trajectory ATE is insensitive to AP-position error out to 32 m in both regimes (localization is odometry-anchored, measurements refinement-only); the map degrades gently and monotonically.

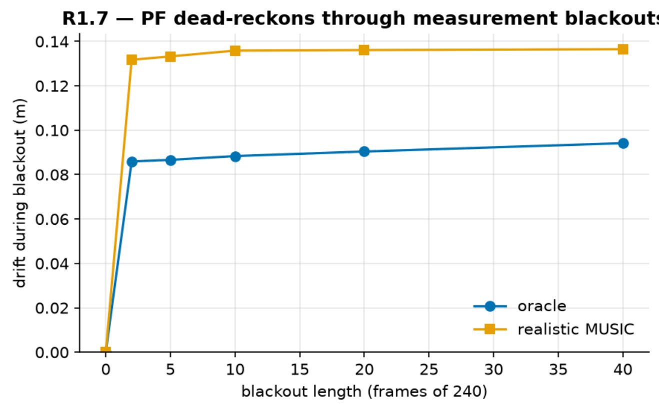

Fig. 7. Blockage robustness. The particle filter dead-reckons through a total measurement blackout: a 40-frame (17% of run) outage adds ≈0.1 m drift and re-locks immediately.

refinement-only; they cannot set absolute position, so AP error cannot corrupt it. The dependence instead lands on the *map*, which degrades gently and monotonically ($\approx +18\%$ across map accuracy, Chamfer, and completeness from 0 to 32 m). And the monostatic geometry (Sec. IX-D) removes the AP dependence outright — it needs no AP position at all.

Sudden measurement loss is absorbed the same way (Fig. 7). A *total* blackout of all detections for up to 40 frames (17% of the run) adds only ≈0.1–0.14 m of drift — nearly constant in blackout length — and the filter re-locks immediately once measurements return. WiFi-SLAM inherits the vehicle's odometry resilience to dropout.

X. Hardware Validation on $30 of Silicon

Everything above is ray-traced. To move the underlying reading onto real hardware, we build the monostatic geometry from two $30 ESP32-S3 boards — one illuminating (HT40, 40 MHz), one logging CSI ≈0.5 m away — and process the CSI with the *same* delay-domain chain, using only the excess delay relative to the line-of-sight arrival (Sec. III-F). Formally this is a bistatic radar-cross-section measurement of a reflector with coherent background subtraction (IEEE Std 1502): recording the channel with and without the reflector and subtracting cancels the receiver's own frequency response, which otherwise manufactures a spurious tap. Working at 2.4 GHz rather than the simulated 5.2 GHz is licensed by the carrier-independence result (Table IV, B→C).

First light. The pipeline runs end-to-end on real silicon. From a static two-board capture we parse valid HT40 records (40 MHz, high-throughput, 384-byte buffers), whose HT-LTF carries 128 subcarriers with exactly 114 non-null — matching the physical active mask (3 DC-notch + 11 guard bins) and so validating the subcarrier ordering. Coherent averaging aligns the line-of-sight tap to 0 m, and with the two boards adjacent the channel is near-flat — a single clean line-of-sight tap, the correct result for that geometry. To our knowledge this is the first delay-resolved reflector reading obtained from an ESP32; nobody has closed this loop at 40 MHz on such hardware before. It demonstrates that the delay-domain reading the whole argument rests on is real, on commodity silicon.

We state the scope precisely. This validates the *reading* — that the delay-domain measurement underpinning the whole argument is obtainable from commodity WiFi silicon — not a full SLAM system. It is a static bench with tape-measured ground truth; we claim no vehicular mapping from it. Measuring the phantom *rate* itself on this hardware, against surveyed reflectors in a large space, is the natural next experiment (Sec. XII); the firmware, the offline chain, and a one-command runner are released with this paper so that it can be reproduced or extended by others at the cost of two development boards.

XI. Discussion and Limitations

The picture is consistent across three sensors and onto real hardware. *Localization* from ambient WiFi is centimetre-level, structurally robust, and one to two orders of magnitude cheaper than the LiDAR/radar it stands in for — a genuinely attractive proposition. *Mapping* is capped by a phantom ceiling that we have shown is a front-end and geometry limit: not path-discrimination (the smallest term), not resolution (immune to bandwidth and aperture), not carrier-specific, and not WiFi-specific. The single most effective lever is geometry — a monostatic, on-vehicle transmitter — which also, conveniently, removes the AP-position dependence entirely.

Several limitations bound the claims and mark the honest edges. The hardware result is a *static* bench, not a moving vehicle; it validates the reading, and the corridor measurement of the phantom rate on a real reflector (Sec. X) is the next data point. Localization's robustness is inseparable from its being *odometry-anchored*: the WiFi measurements refine but do not carry absolute position, which is why they tolerate AP error and dropout, and also why they cannot rescue a failed odometry. The simulation, though ray-traced and anchored to real LiDAR on one arm, is still simulation. And the single-ULA front-end cannot observe arrival elevation, which both bounds the achievable discrimination and was the source of an over-optimistic earlier estimate.

XII. Future Work

Three lines follow directly. First, **measure the phantom rate on the hardware**: with the bench extended into a large surveyed space, the same detection chain yields the physical counterpart of Table IV — MUSIC versus CFAR and bistatic versus monostatic measured on real silicon rather than inferred from it. Our own results predict the ceiling will reproduce; the discriminating quantities are the *differences* between those front-ends and geometries, not the absolute number. Second, **port the winning chain on-chip**: the monostatic delay-domain detector fits an ESP32-S3 with roughly two orders of magnitude of headroom (Sec. III-D), which would make the sensor self-contained rather than tethered to an offline host. Third, and most open, **a localization front-end that does not lean on odometry**: the robustness reported in Sec. IX-E is real but inherited, and a WiFi measurement model able to carry absolute position on its own — rather than refine a motion estimate — would change what the modality can claim.

XIII. Conclusion

Ambient WiFi is a practical, inexpensive localization sensor for automotive SLAM and a poor mapping sensor — and the reason it maps poorly is now specific and quantified. Most realistic-CSI detections are phantoms, and a several-metre range bias corrupts the rest; this is a property of the superresolution front-end and the bistatic geometry, not of the carrier, the bandwidth, or WiFi as such, as a controlled comparison against LiDAR and a 77 GHz radar on one substrate shows. The open problem for WiFi mapping is therefore the front-end, not the physics, and the clearest escape is a monostatic on-vehicle geometry, which nearly eliminates the phantoms and needs no known transmitter positions. We have shown the delay-domain reading holds on $30 of real ESP32 silicon; measuring the phantom rate itself on that hardware, and porting the winning chain on-chip, are the immediate next steps.

Reproducibility

All code, configs, and result data are in the repository above (AGPL-3.0); the simulation is Sionna RT; the hardware pipeline and firmware are included.

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